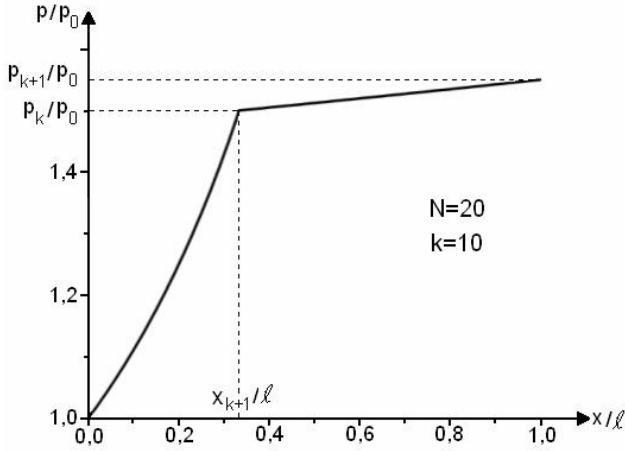


Marking scheme
Points are given for any correct solution

Problem II
Bicycle pump

Nr. item	Task No. 1	Point
1.a.	For: expression of number of moles of air, initially found in the tire at atmospheric pressure p_0 and at T_0 $v_{initial} = \frac{p_0 \cdot V_r}{R \cdot T_0}$ 0.20p	0.80p
	expression of number of moles of air pumped inside the tire at each pumping cycle. $\begin{cases} v_0 = \frac{p_0 \cdot V_p}{R \cdot T_0} \\ v_0 = \frac{p_0 \cdot V_r}{R \cdot T_0 \cdot N} \end{cases}$ 0.20p	
	expression of the number of moles of air, v_k that are within the tire after the student pumped air k times using the pump with the hand operated piston $v_k = v_{initial} + k \cdot v_0$ 0.20p	
	$v_k = \frac{p_0 \cdot V_r}{R \cdot T_0} \cdot \left(1 + \frac{k}{N}\right)$ 0.20p	
1.b.	For: expression of the air pressure p_k within the tire, after the student pumped air k times $p_k \cdot V_r = v_k \cdot R \cdot T_0$ 0.20p	0.40p
	$p_k = p_0 \cdot \frac{N+k}{N}$ 0.20p	
1.c.	For: isothermal transformation law applied to the air found in the bicycle pump $p_0 \cdot \frac{V_r}{N} = p_k \cdot S \cdot (\ell - x_{k+1})$ 0.20p	0.40p
	$x_{k+1} = \ell \cdot \frac{k}{N+k}$ 0.20p	

1.d.	For: $p_0 \cdot \frac{V_r}{N} = p(x) \cdot \frac{V_r}{N \cdot \ell} \cdot (\ell - x), \text{ for } 0 \leq x \leq \frac{k \cdot \ell}{N + k} \quad 0.20p$ $p(x) \cdot \left[V_r + (\ell - x) \cdot \frac{V_r}{N \cdot \ell} \right] = p_{k+1} \cdot V_r, \text{ for } \frac{k \cdot \ell}{N + k} < x \leq \ell \quad 0.20p$ $p(x) = \begin{cases} \frac{p_0}{\left(1 - \frac{x}{\ell}\right)}, & 0 \leq x \leq \frac{k \cdot \ell}{N + k} \\ \frac{p_0 \cdot (N + k + 1)}{\left[(N + 1) - \frac{x}{\ell}\right]}, & \frac{k \cdot \ell}{N + k} < x \leq \ell \end{cases} \quad 0.20p$	0.60p
1.e.	For: $p'(x) = \begin{cases} \frac{p_0/\ell}{\left(1 - \frac{x}{\ell}\right)^2}, & 0 \leq x < \frac{k \cdot \ell}{N + k} \\ \frac{p_0 \cdot (N + k + 1)/\ell}{\left[(N + 1) - \frac{x}{\ell}\right]^2}, & \frac{k \cdot \ell}{N + k} < x \leq \ell \end{cases} \quad 0.20p$ <i>Observation: First-order derivative of the function $p(x)$ is positive over the entire range of definition</i> $\begin{cases} p'_s(x_k) = \frac{p_0/\ell}{\left(1 - \frac{k}{N + k}\right)^2} = \frac{p_0 \cdot (N + k)^2}{\ell \cdot N^2} \\ p'_d(x_k) = \frac{p_0 \cdot (N + k + 1)/\ell}{\left[(N + 1) - \frac{x}{\ell}\right]^2} = \frac{p_0 \cdot (N + k)^2}{\ell \cdot N^2 \cdot (N + k + 1)} \end{cases} \quad 0.20p$ <i>Observation: Left and right of the point where the valve opens, the pressure shows an angular point.</i> $p''(x) = \begin{cases} \frac{2p_0/\ell^2}{\left(1 - \frac{x}{\ell}\right)^3}, & 0 \leq x < \frac{k \cdot \ell}{N + k} \\ \frac{2p_0 \cdot (N + k + 1)/\ell^2}{\left[(N + 1) - \frac{x}{\ell}\right]^3}, & \frac{k \cdot \ell}{N + k} < x \leq \ell \end{cases} \quad 0.20p$ <i>Observation: Second-order derivative is also positive on the whole range of definition.</i>	0.80p

	 0.20p Observation: Graphic representation refers to the 11-th pumping cycle for $N = 20$	
Nr. item	Task No. 2	Points
2.a.	For: $p_{k_0-1} < n \cdot p_0 \leq p_{k_0}$ 0.20p $p_{k_0} = p_0 \cdot \frac{N + k_0}{N}$ 0.20p $k_0 \geq N \cdot (n - 1)$, k_0 number of pumping cycle is a natural number 0.20p $k_0 = N \cdot (n - 1)$, if $N \cdot (n - 1)$ is a natural number or $k_0 = [N \cdot (n - 1)] + 1$, if $N \cdot (n - 1)$ is not a natural number 0.60p	1.20p
2.b.	For: find the work that the student needs to do on the ν_1 mole of air, evolving isothermally from pressure p_0 to pressure p_{k-1} 0.20p $L_{K,I} = \nu_1 \cdot R \cdot T_0 \cdot \ln \frac{p_{k-1}}{p_0}$ $L_{K,I} = \frac{p_0 \cdot V_r}{N} \cdot \ln \frac{N + k - 1}{N}$ 0.20p expression of the work that the student needs to do on the ν_k mole of air, evolving isothermally from pressure p_{k-1} to pressure p_k 0.20p $L_{K,II} = \nu_k \cdot R \cdot T_0 \cdot \ln \frac{p_k}{p_{k-1}}$	1.20p

	$L_{K, II} = \frac{p_0 \cdot V_r \cdot (N+k)}{N} \cdot \ln \frac{N+k}{N+k-1}$ expression of the total work that the student needs to do, during the k pumping cycle. $\begin{cases} L_{K, total} = L_{K, I} + L_{K, II} \\ L_{K, total} = \frac{p_0 \cdot V_r}{N} \cdot \ln \frac{N+k-1}{N} + \frac{p_0 \cdot V_r \cdot (N+k)}{N} \cdot \ln \frac{N+k}{N+k-1} \\ L_{K, total} = \frac{p_0 \cdot V_r}{N} \cdot \left[\ln \frac{N+k-1}{N} + (N+k) \cdot \ln \frac{N+k}{N+k-1} \right] \end{cases}$ expression of the total work that the student do from the moment he starts pumping till the moment when the pressure inside tire is $n \cdot p_0$ $L_{total} = \frac{p_0 \cdot V_r}{N} \cdot \sum_{k=1}^{N \cdot (n-1)} \left[\ln \frac{N+k-1}{N} + (N+k) \cdot \ln \frac{N+k}{N+k-1} \right]$	0.20p	
Nr. item	Task No. 3	Points	
3.a.	For: number of air moles within the tire after $k = 10$ pumping cycles $\nu_{10} = 0,43 \text{ moli}$	0.20p	0.20p
3.b.	For: value of the air pressure within the tire after 10 pumping cycles $p_{10} = 1,52 \cdot 10^5 \frac{N}{m^2}$	0.20p	0.20p
3.c.	For: number of cycles needed for the air pressure within the tire to reach the value $n \cdot p_0 \quad \begin{cases} k_0 = [20 \cdot (2,51 - 1)] + 1 \\ k_0 = 31 \end{cases}$	0.20p	0.20p
3.d.	For: value of the work that the student has to do during the 10th pumping cycle $\begin{cases} L_{10, total} = \frac{\left(1,01 \cdot 10^5 \frac{N}{m^2} \right) \cdot (7,00 \cdot 10^{-3} m^3)}{20} \cdot \left[\ln \frac{29}{20} + 30 \cdot \ln \frac{30}{29} \right] \\ L_{10, total} \cong 49,1 J \end{cases}$	0.20p	0.20p

Nr. item	Task No. 4	Points
4.a.	For: expression of partial pressure $p_{aer}(T_0)$ of air inside the tire $\begin{cases} p_{aer}(T_0) = p - p_{s,0} \\ p_{aer}(T_0) = 2,477 \cdot p_0 \end{cases}$ 0.20p	0.80p
	$\begin{cases} p_{aer}(T_1) = p_{aer}(T_0) \cdot \frac{T_1}{T_0} \\ p_{aer}(T_1) = 2,229 \cdot p_0 \end{cases}$ 0.20p	
	$\begin{cases} p_1 = p_{aer}(T_1) + p_{s,1} \\ p_1 = 2,235 \cdot p_0 \end{cases}$ 0.20p	
	The value of the pressure inside the tire in specified conditions of use $p_1 = 2,257 \times 10^5 \text{ N} \cdot \text{m}^{-2}$ 0.20p	
4.b.	For : expression of the concentration n_e^* of nitrogen molecules in the room where the tire is kept $\begin{cases} n_e^* = \frac{p_0}{T_0} \cdot \frac{1}{k_B} \\ n_e^* = \frac{p_0}{T_0} \cdot \frac{N_A}{R} \end{cases}$ 0.20p	1.20p
	expression of the concentration n_i^* of nitrogen molecules in the tire $\begin{cases} n_i^* = 2,51 \cdot \frac{p_0}{T_0} \cdot \frac{N_A}{R} \\ n_i^* = 2,51 \cdot \frac{p_0}{T_0} \cdot \frac{1}{k_B} \end{cases}$ 0.20p	
	expression of the density of the flux of molecules N_e^* coming from all directions, from outside the tire $\begin{cases} N_e^* = \frac{1}{4} \cdot n_e^* \cdot \bar{v} \\ N_e^* = \frac{1}{4} \cdot \frac{p_0}{T_0} \cdot \frac{N_A}{R} \cdot \sqrt{\frac{8R \cdot T_0}{\pi \cdot \mu_{azot}}} \end{cases}$ 0.20p	

	expression of the density of the flux of molecules N_i^* coming from all directions, from inside the tire $\begin{cases} N_i^* = 2,51 \cdot \frac{1}{4} \cdot n_i^* \cdot \bar{v} \\ N_i^* = 2,51 \cdot \frac{1}{4} \cdot \frac{p_0}{T_0} \cdot \frac{N_A}{R} \cdot \sqrt{\frac{8R \cdot T_0}{\pi \cdot \mu_{azot}}} \end{cases}$ 0.20p	
	expression of the number N_i of nitrogen molecules falling onto the surface of pinhole in unit of time coming from all directions, from inside the tire $N_i = 2,51 \cdot \frac{1}{4} \cdot \frac{p_0}{T_0} \cdot \frac{N_A}{R} \cdot S \cdot \sqrt{\frac{8R \cdot T_0}{\pi \cdot \mu_{azot}}} \quad N_i = 2,51 \cdot p_0 \cdot N_A \cdot S \cdot \sqrt{\frac{1}{2\pi \cdot \mu_{azot} \cdot R \cdot T_0}}$ 0.20p	
	expression of the number N_e of nitrogen molecules falling onto the surface of pinhole in unit of time coming from all directions, from outside the tire $N_e = \frac{1}{4} \cdot \frac{p_0}{T_0} \cdot \frac{N_A}{R} \cdot S \cdot \sqrt{\frac{8R \cdot T_0}{\pi \cdot \mu_{azot}}} \quad N_e = p_0 \cdot N_A \cdot S \cdot \sqrt{\frac{1}{2\pi \cdot \mu_{azot} \cdot R \cdot T_0}}$ 0.20p	
4.c.	For: expression of the number of nitrogen molecules falling onto the surface S of the small orifice in time dt coming from outside the tire $\begin{cases} \mathfrak{N}_e = S \cdot N_e^* \cdot dt \\ \mathfrak{N}_e = \frac{1}{4} \cdot \frac{p_0}{T_0} \cdot \frac{N_A}{R} \cdot \bar{v} \cdot S \cdot dt \end{cases}$ 0.20p expression of the number of nitrogen molecules falling onto the surface S of the small orifice in time dt coming from inside the tire depends on pressure $\begin{cases} \mathfrak{N}_i = \frac{1}{4} \cdot n^*(p) \cdot \bar{v} \cdot S \cdot dt \\ \mathfrak{N}_i = \frac{1}{4} \cdot \frac{p}{T_0} \cdot \frac{N_A}{R} \cdot \bar{v} \cdot S \cdot dt \end{cases}$ 0.20p number of molecules inside the tire $\mathfrak{N} = \frac{p \cdot V_r}{R \cdot T_0} \cdot N_A$ 0.20p expression of the variation of numbers of molecules inside the tire $\begin{cases} d\mathfrak{N} = \mathfrak{N}(t+dt) - \mathfrak{N}(t) \\ d\mathfrak{N} = \frac{V_r \cdot N_A}{R \cdot T_0} \cdot (p(t+dt) - p(t)) \\ d\mathfrak{N} = \frac{V_r \cdot N_A}{R \cdot T_0} \cdot dp \end{cases}$ 0.20p $d\mathfrak{N} = \mathfrak{N}_e - \mathfrak{N}_i$ 0.20p	1.80p

$\frac{V_r \cdot N_A}{R \cdot T_0} \cdot dp = \frac{1}{4} \cdot \frac{p_0}{T_0} \cdot \frac{N_A}{R} \cdot \bar{v} \cdot S \cdot dt - \frac{1}{4} \cdot \frac{p}{T_0} \cdot \frac{N_A}{R} \cdot \bar{v} \cdot S \cdot dt$	0.20p
$\frac{dp}{(p - p_0)} = -\frac{1}{4} \cdot \frac{\bar{v} \cdot S}{V_r} \cdot dt$	0.20p
$\int_{2,51 \cdot p_0}^{1,10 \cdot p_0} \frac{dp}{(p - p_0)} = -\frac{1}{4} \cdot \frac{\bar{v} \cdot S}{V_r} \cdot \int_0^{t_f} dt \quad t_f = \frac{4 \cdot V_r}{\bar{v} \cdot S} \cdot \ln 15,1$	0.20p
$\begin{cases} t_f = 1,59 \times 10^4 \text{ s} \\ t_f \cong 4,4 \text{ ore} \end{cases}$	0.20p
TOTAL Problem II	
10p	

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